PRE-ANALYSIS PLAN: #MANYDAUGHTERS

RT: Carmen van Klaveren and Egon Tripodi

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1: General Exclusions

- Private households
- Non-heterosexual people (i.e., keep if sexor==0).
- People with no children (i.e., keep if biovalid==31).
- People with more than three children.
- Cases in which the two parents report different number of children.
- People who did not have a successful SOEP interview (i.e., keep if nett1==1).
- Our analyses are run in the last cross-sectional wave of the dataset, which for some hypotheses is 2018 (H1, H3, H4) and for others 2019 (H2). The survey wave selection is based on data availability and quality.

2: Variables

Individual has at least one daughter (d_daughter)

- We calculated the number of daughters based on the variables “kidsex” and generated a binary variable taking the value 1 if the number of daughters is greater than or equal to one.

Number of children (sumkids)

- As given in the SOEP.

Attitude towards gender equality (plh0308_v2)

- Level of agreement (1 = Totally disagree; 7 = Totally agree) with statement “It is best if man and woman work the same amount so they can share the responsibility in housework and childcare”.

Political attitude (plh0004)

- Indication of political orientation, where value 0 is extreme right-leaning and value 10 is extreme left-leaning.
- This variable was reverse coded to fit hypothesis framing (higher number = left-leaning political attitudes).

Behavior indicative of gender equality (equality_fin_dec)

- Binary variable taking value of 1 if both partners make financial decisions equally and 0 otherwise.
- This variable is constructed from variable plc0011, which indicates who takes financial decisions in the household (1 = oneself; 2 = partner; 3 = both partners equally). The binary variable takes a value of 1 if plc0011 = 3 (both partners decide equally) and value 0 otherwise.

Controls:
- Sex (sex): Takes value of 1 if male, 2 if female.
- Born in Germany (germborn): Takes value of 1 if born in Germany, 0 if not born in Germany.
- Age: We deducted the year of birth (gebjahr) from 2025 (current year).
- Age^2: Squared variable age (above).

3: Other Considerations

We shall conduct a similar test to Washington (2008), to rule out the predominance of son preference in this sample. With the sample restrictions we implemented, especially on the number of children in the sample, we don't find in this 5% of the sample that the gender of the first born determines how many children people choose to have.

Reference:
Washington, Ebonya L. 2008. "Female Socialization: How Daughters Affect Their Legislator Fathers' Voting on Women's Issues." *American Economic Review* 98 (1): 311–32.

4: Analysis

Hypothesis 1:

Having daughters leads to stronger support for gender equality (as measured by attitudes).
That is, denoting X as a measure of support for gender equality,

(X for respondents with daughters) > (X for respondents without daughters)

We estimate the following regression model (for 2018):

$Y_i = \alpha + \beta_1 d_daughter_i + \beta_2 X_i + \delta_{sumkids} + \varepsilon_i$

where:

- Y_i is the individual's attitude towards gender equality (plh0308_v2), stating their level of agreement (1 = Totally disagree; 7 = Totally agree) with the statement: "It is best if man and woman work the same amount so they can share the responsibility in housework and childcare".
- d_daughter is an indicator for whether the individual has any daughter, taking the value 1 if the number of daughters is greater than or equal to one.
- sumkids is the number of children, where every distinct realization is included in the regression with a dummy (i.e. via i.sumkids).
- X_i is the set of controls (age, age^2, sex, born in Germany).

Standard errors are clustered at the household level.

Test and Interpretation: We then test whether the coefficient for d_daughter is positive (using a t test). If this coefficient is positive and if the null hypothesis is rejected, we interpret this result as supporting hypothesis 1.

Hypothesis 2:

Having daughters leads to more left-leaning political preferences.
That is, denoting Y as a measure of left-leaning political preferences,

(Y for respondents with daughters) > (Y for respondents without daughters)

We estimate the following regression model (for 2019):

$Y_i = \alpha + \beta_1 d_daughter_i + \beta_2 X_i + \delta_{sumkids} + \varepsilon_i$
where,

- Y_i is the individual's political orientation (plh0004), where value 0 is extreme right-leaning and value 10 is extreme left-leaning.
- d_daughter is an indicator for whether the individual has any daughter, taking the value 1 if the number of daughters is greater than or equal to one.
- sumkids is the number of children, where every distinct realization is included in the regression with a dummy (i.e. via i.sumkids).
- X_i is the set of controls (age, age^2, sex, born in Germany).

Standard errors are clustered at the household level.

Test and Interpretation: We then test whether the coefficient for d_daughter is positive (using a t test). If this coefficient is positive and if the null hypothesis is rejected, we interpret this result as supporting hypothesis 2.

<u>Hypothesis 3</u>
Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).
That is, denoting Z as a measure of gender equality in the household,

(Z for respondents with daughters) > (Z for respondents without daughters)

We estimate the following regression model (for 2018):
$Y_i = \alpha + \beta_1 d_daughter_i + \beta_2 X_i + \delta_{sumkids} + \varepsilon_i$
where,

- Y_i is a binary variable (equality_fin_dec) taking value of 1 if both partners make financial decisions equally in the household, and 0 otherwise.
- d_daughter is an indicator for whether the individual has any daughter, taking the value 1 if the number of daughters is greater than or equal to one.
- sumkids is the number of children, where every distinct realization is included in the regression with a dummy (i.e. via i.sumkids).
- X_i is the set of controls (age, age^2, sex, born in Germany).

Standard errors are clustered at the household level.

Test and Interpretation: We test whether the coefficient for d_daughter is positive (using a t test). If this coefficient is positive and if the null hypothesis is rejected, we interpret this result as supporting hypothesis 3.

<u>Hypothesis 4:</u>
Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).
That is, denoting X as a measure of support for gender equality,

(X for male respondents with daughters − X for male respondents without daughters)
> (X for female respondents with daughters − X for female respondents without daughters)

We estimate the following regression model separately for men and women in the sample (for 2018):
$Y_i = \alpha + \beta_1 d_daughter_i + \beta_2 ib2.sex_i + \beta_3\ d_daughter_i * ib2.sex_i + \beta_4 X_i + \delta_{sumkids} + \varepsilon_i$

where,

- Y_i is the individual's attitude towards gender equality (plh0308_v2), stating their level of agreement (1 = Totally disagree; 7 = Totally agree) with the statement: "It is best if man and woman work the same amount so they can share the responsibility in housework and childcare".
- d_daughter is an indicator for whether the individual has any daughter, taking the value 1 if the number of daughters is greater than or equal to one.
- ib2.sex is a sex dummy (where being female is the baseline category).
- sumkids is the number of children, where every distinct realization is included in the regression with a dummy (i.e. via i.sumkids).
- X_i is the set of controls (age, age^2, born in Germany).

Heteroskedasticity robust standard errors.

Test and Interpretation: We then test whether the coefficient for the interaction of d_daughter and ib2.sex is positive (using a t test). If this coefficient is positive and if the null hypothesis is rejected, we interpret this result as supporting hypothesis 4.